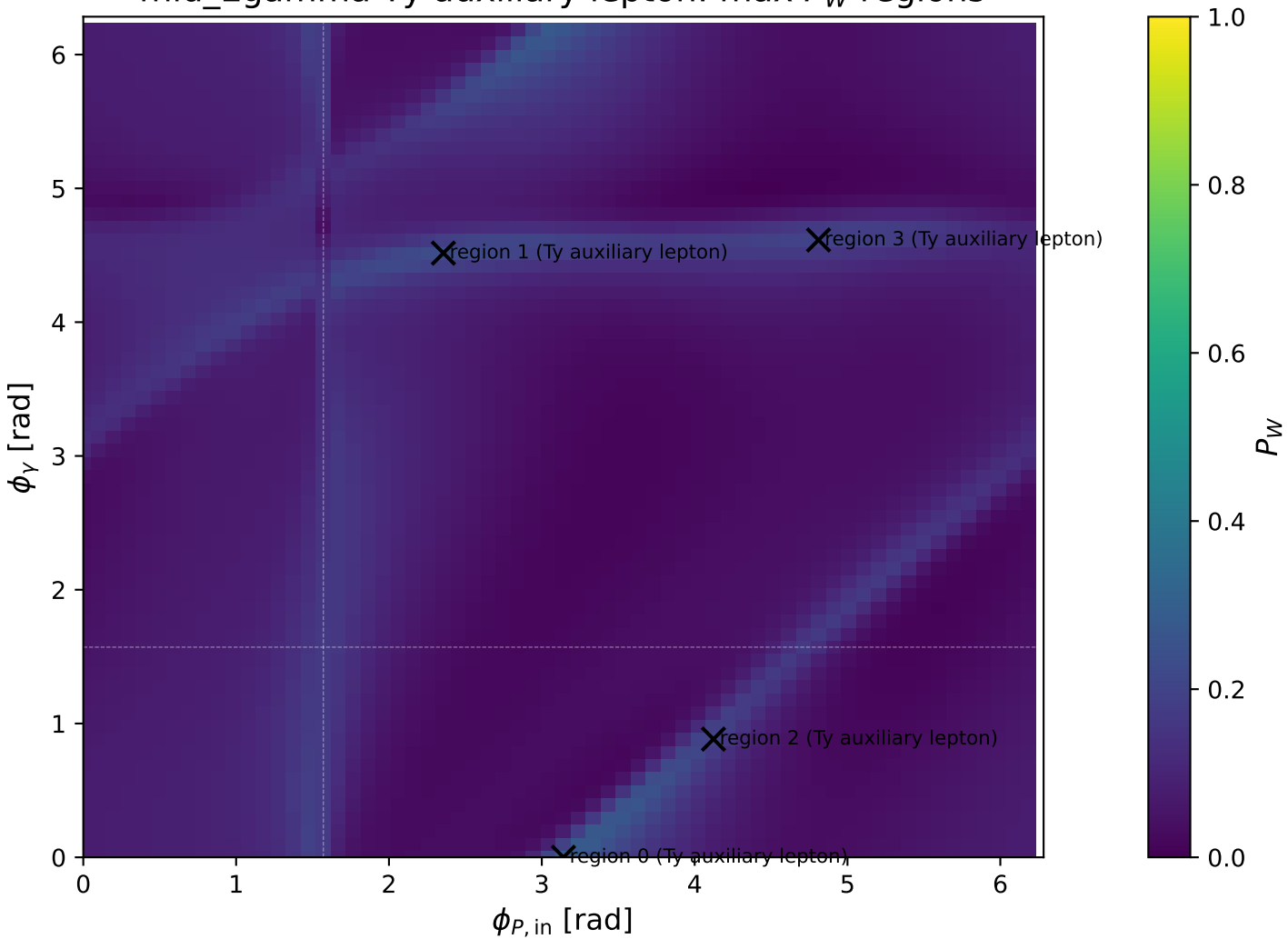
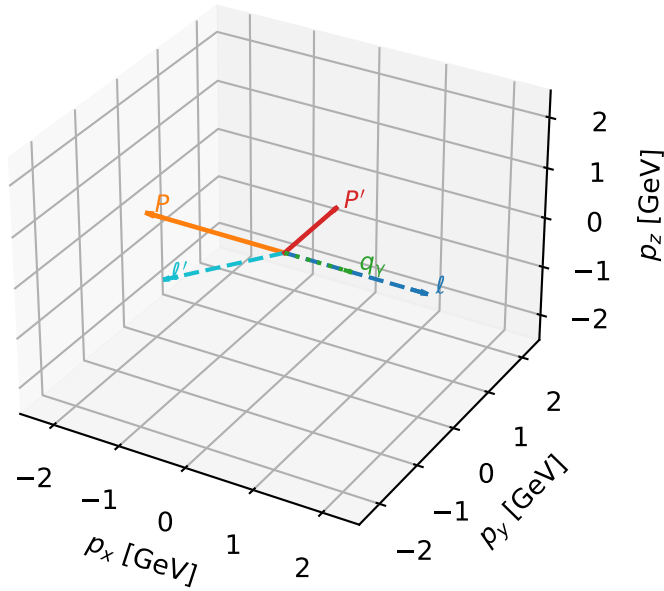


mid\_Egamma Ty auxiliary lepton: max  $P_W$  regions



# Ty auxiliary lepton characteristic max $P_W$ configuration

3D momenta



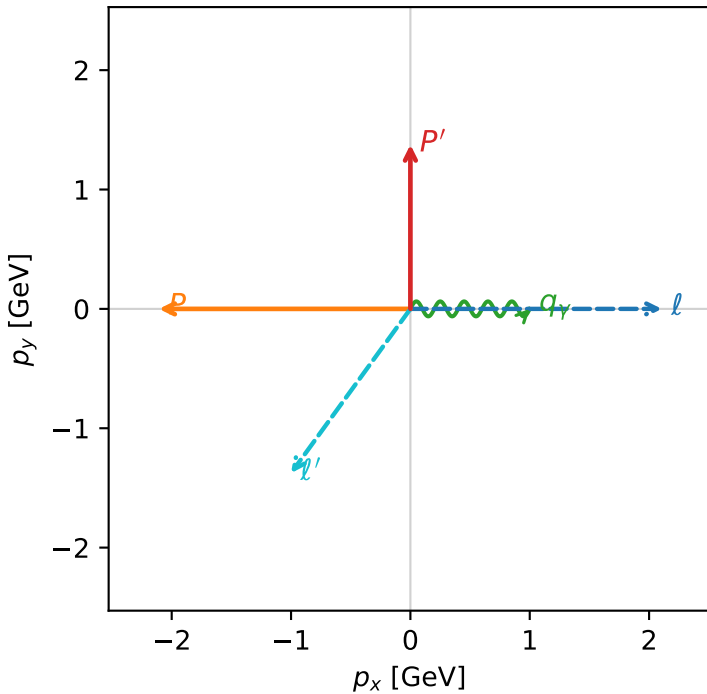
w\_purity\_mid\_egamma\_ty\_lepton\_region\_0 P\_W=0.277093  
 region: mid\_Egamma\_Ty\_lepton\_region\_0  
 outgoing-state purity: 0.551947  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $\frac{1}{2} \sum_{h_p} \rho(T_{y,l}, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.37604$   
 $\theta_{in}=1.57079$ ,  $\phi_l=0$ ,  $\phi_P=3.14159$   
 $E_V=1$ ,  $\phi_Y=0$   
 $Q^2=11.4177$ ,  $x_B=0.774167$ ,  $t=-5.92183$ ,  $W^2=4.2105$ ,  $y=0.751086$   
 $\theta(l', \gamma)=126.007$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$l$   $E= 2.3322$   $p_x= 2.1069$   $p_y= -0.0000$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= -2.1069$   $p_y= 0.0000$   $p_z= 0.0000$   
 $l'$   $E= 1.9732$   $p_x= -1.0000$   $p_y= -1.3760$   $p_z= 0.0000$   
 $P'$   $E= 1.6653$   $p_x= 0.0000$   $p_y= 1.3760$   $p_z= 0.0000$   
 $q_Y$   $E= 1.0000$   $p_x= 1.0000$   $p_y= 0.0000$   $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=-1, h_Y=+1$   
 electron Ty, proton h=-1

$h_e=+1, h_p=+1, h_Y=-1$   
 electron Ty, proton h=+1

$h_e=-1, h_p=+1, h_Y=+1$   
 electron Ty, proton h=-1

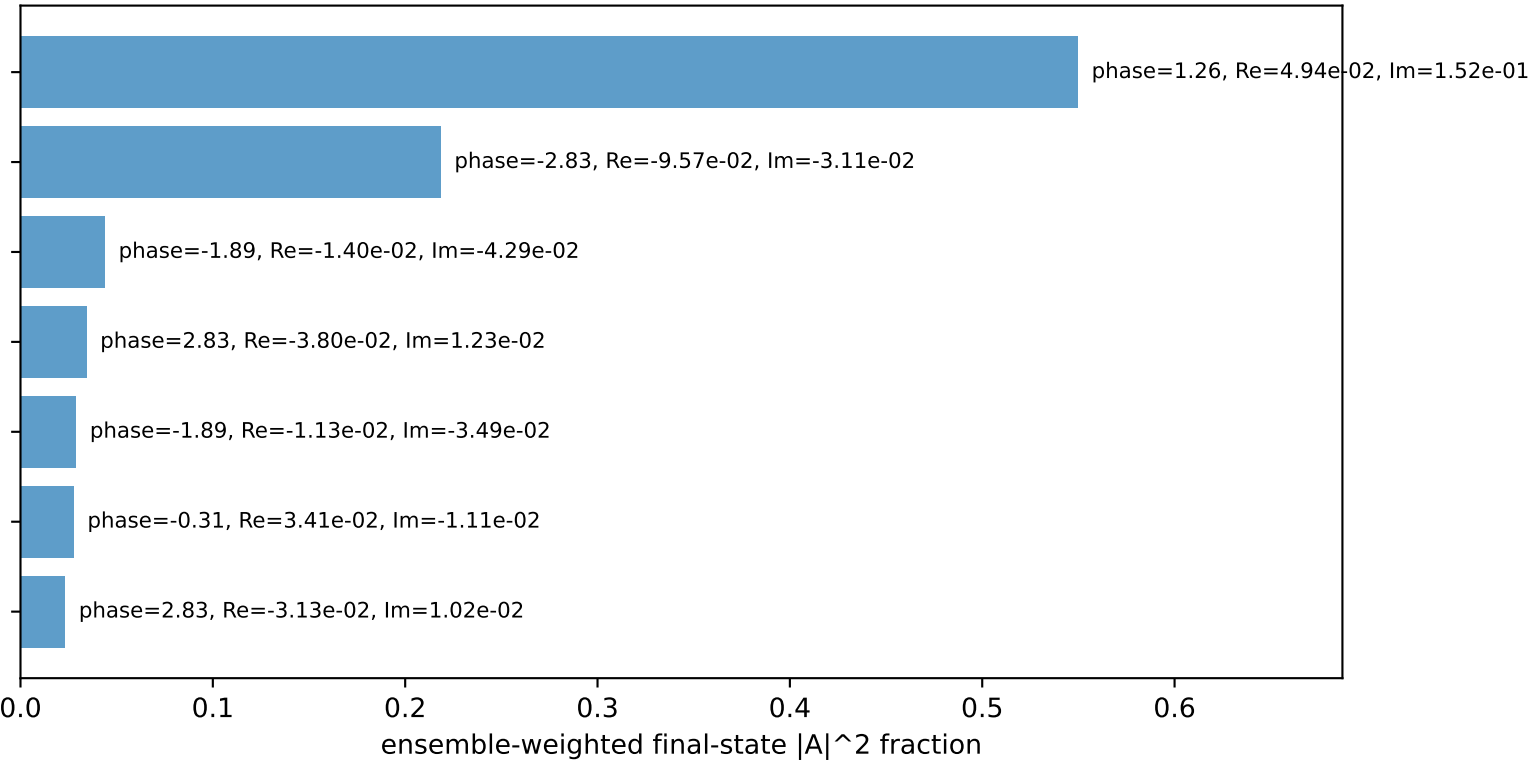
$h_e=-1, h_p=+1, h_Y=+1$   
 electron Ty, proton h=+1

$h_e=-1, h_p=-1, h_Y=-1$   
 electron Ty, proton h=-1

$h_e=-1, h_p=+1, h_Y=-1$   
 electron Ty, proton h=+1

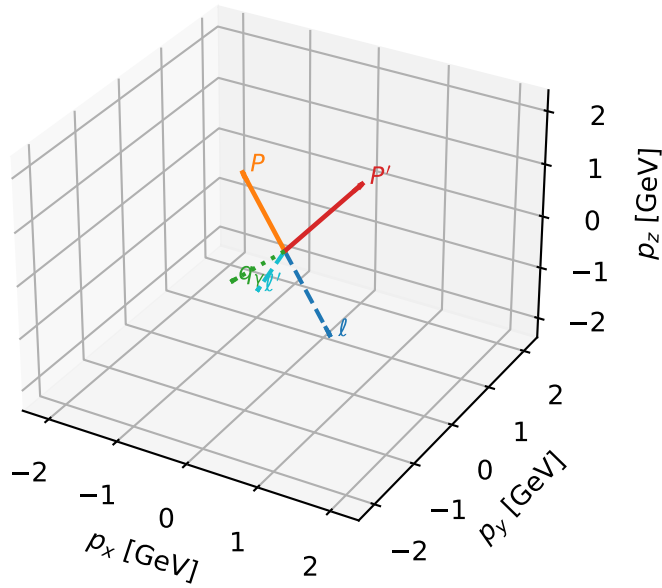
$h_e=-1, h_p=-1, h_Y=+1$   
 electron Ty, proton h=+1

Leading initial-ensemble/final-state components (N=7, retained=92.7%)



# Ty auxiliary lepton characteristic max $P_W$ configuration

3D momenta



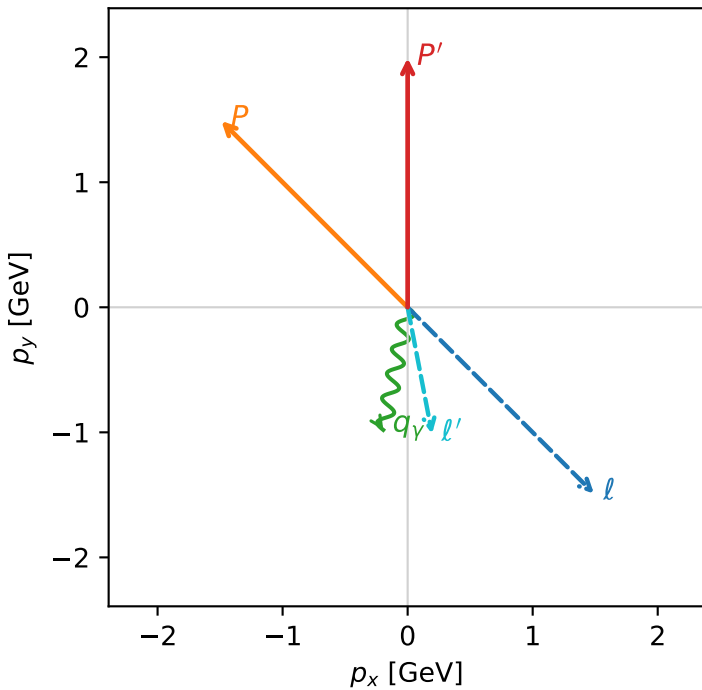
w\_purity\_mid\_egamma\_ty\_lepton\_region\_1 P\_W=0.23088  
 region: mid\_Egamma\_Ty\_lepton\_region\_1  
 outgoing-state purity: 0.523181  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(T_{y,l},h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.99277$   
 $\theta_{in}=1.57079$ ,  $\phi_l=5.49779$ ,  $\phi_p=2.35619$   
 $E_\gamma=1$ ,  $\phi_\gamma=4.51604$   
 $Q^2=1.10155$ ,  $x_B=0.116994$ ,  $t=-2.46177$ ,  $W^2=9.19378$ ,  $y=0.479502$   
 $\theta(l',\gamma)=22.1616$  deg

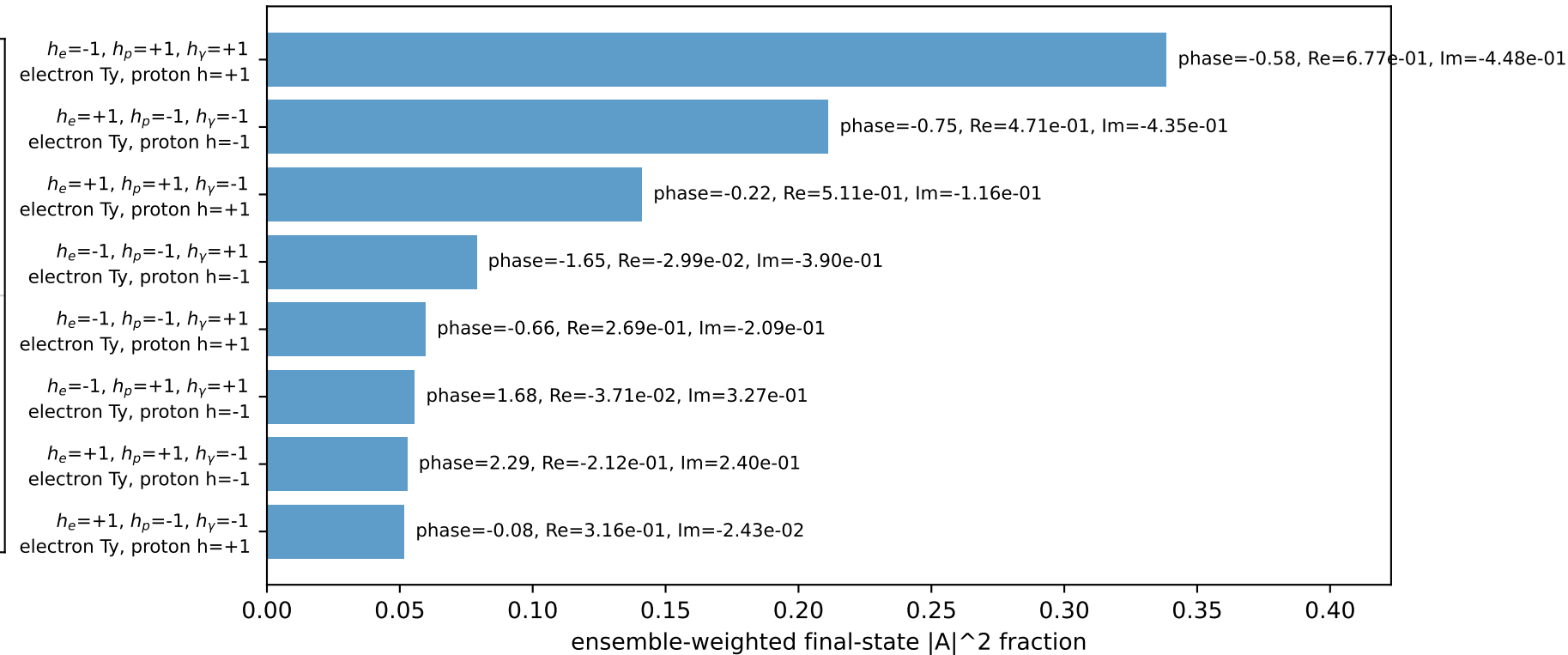
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$l$   $E= 2.3322$   $p_x= 1.4898$   $p_y= -1.4898$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= -1.4898$   $p_y= 1.4898$   $p_z= 0.0000$   
 $l'$   $E= 1.4360$   $p_x= 0.1951$   $p_y= -1.0120$   $p_z= 0.0000$   
 $P'$   $E= 2.2025$   $p_x= 0.0000$   $p_y= 1.9928$   $p_z= 0.0000$   
 $q_Y$   $E= 1.0000$   $p_x= -0.1951$   $p_y= -0.9808$   $p_z= 0.0000$

Transverse plane

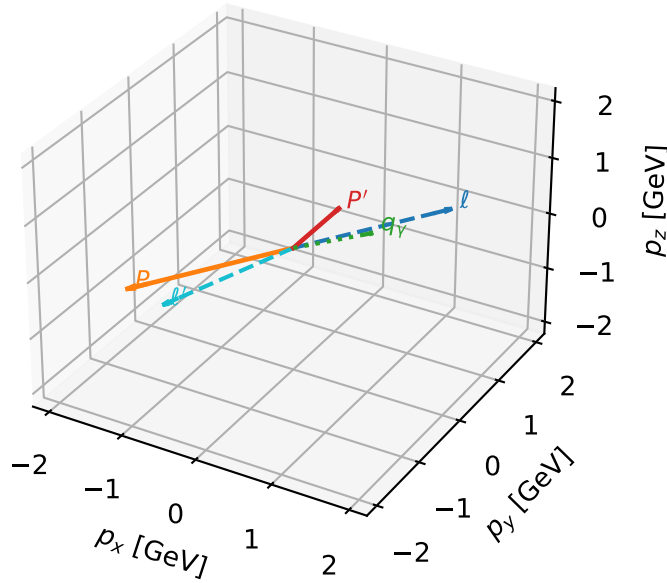


Leading initial-ensemble/final-state components (N=8, retained=98.9%)



# Ty auxiliary lepton characteristic max $P_W$ configuration

3D momenta



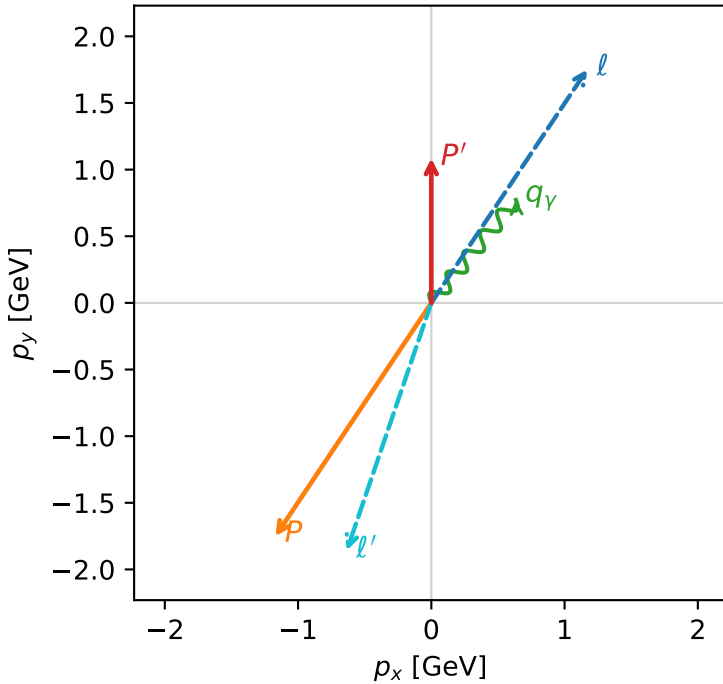
w\_purity\_mid\_egamma\_ty\_lepton\_region\_2 P\_W=0.197956  
 region: mid\_Egamma\_Ty\_lepton\_region\_2  
 outgoing-state purity: 0.52992  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $\frac{1}{2} \sum_{h_p} \rho(T_{y,l}, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.08559$   
 $\theta_{in}=1.57079$ ,  $\phi_l=0.981748$ ,  $\phi_p=4.12334$   
 $E_\gamma=1$ ,  $\phi_\gamma=0.883573$   
 $Q^2=16.2768$ ,  $x_B=0.931815$ ,  $t=-8.66159$ ,  $W^2=2.07089$ ,  $y=0.88958$   
 $\theta(l', \gamma)=159.471$  deg

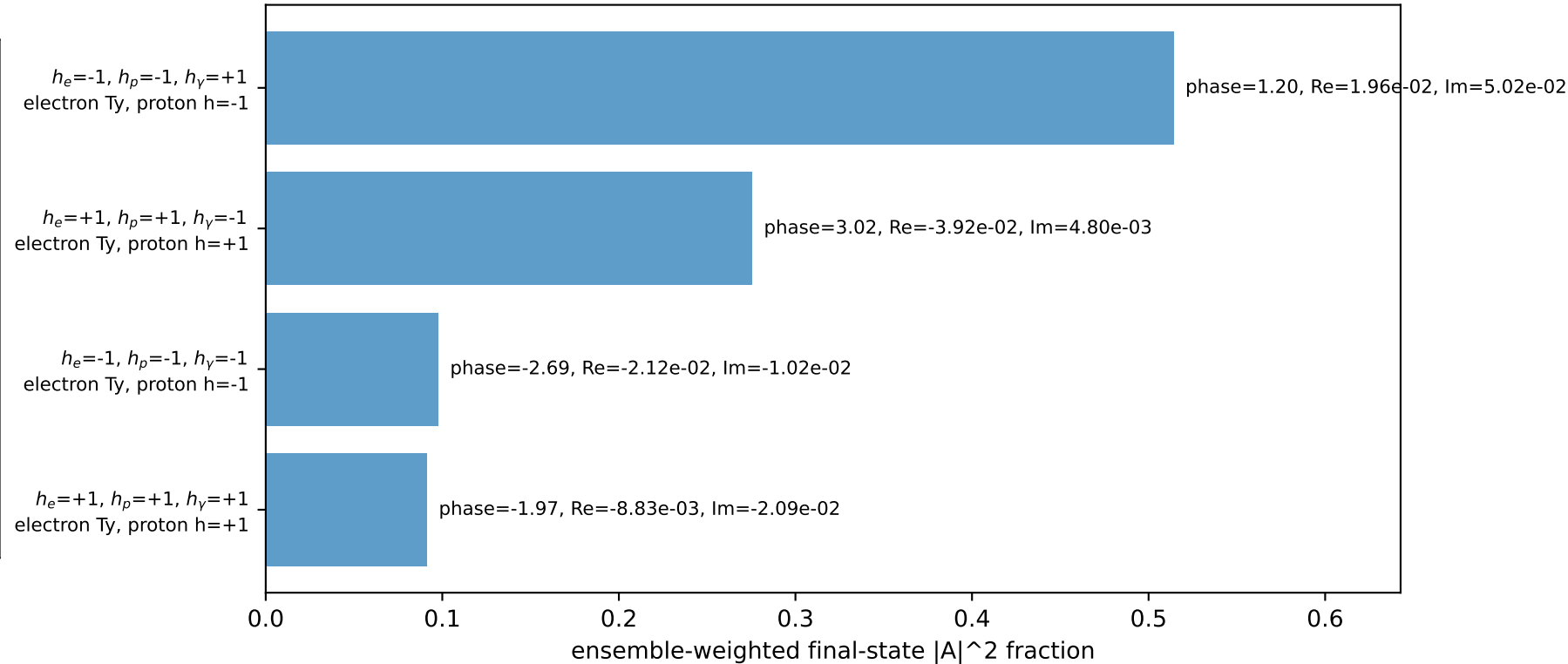
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$l$   $E= 2.3322$   $p_x= 1.1706$   $p_y= 1.7519$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= -1.1706$   $p_y= -1.7519$   $p_z= 0.0000$   
 $l'$   $E= 2.2038$   $p_x= -0.6344$   $p_y= -1.8586$   $p_z= 0.0000$   
 $P'$   $E= 1.4347$   $p_x= 0.0000$   $p_y= 1.0856$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= 0.6344$   $p_y= 0.7730$   $p_z= 0.0000$

Transverse plane

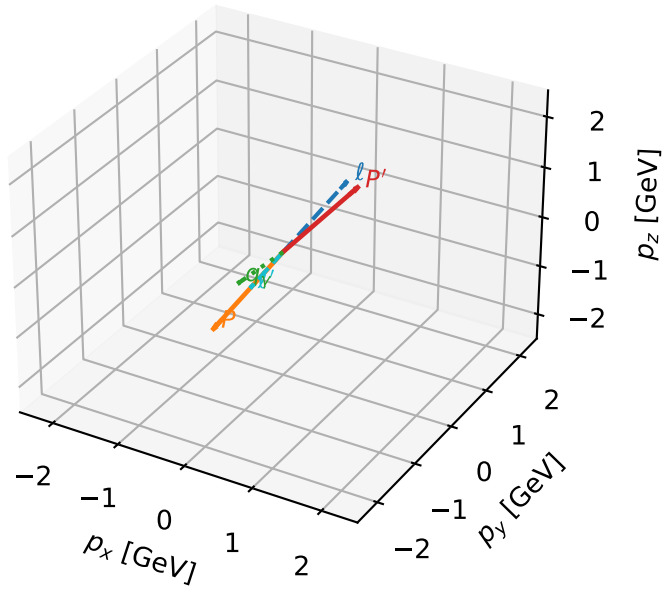


Leading initial-ensemble/final-state components (N=4, retained=97.9%)



# Ty auxiliary lepton characteristic max $P_W$ configuration

3D momenta



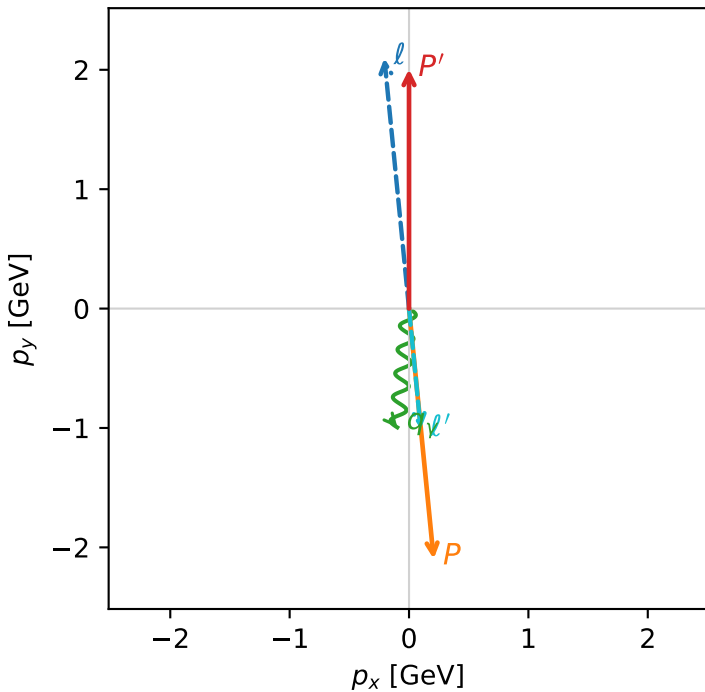
w\_purity\_mid\_egamma\_ty\_lepton\_region\_3 P\_W=0.19767  
 region: mid\_Egamma\_Ty\_lepton\_region\_3  
 outgoing-state purity: 0.500029  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(T_{y,l},h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=2.00525$   
 $\theta_{in}=1.57079$ ,  $\phi_l=1.66897$ ,  $\phi_p=4.81056$   
 $E_V=1$ ,  $\phi_V=4.61421$   
 $Q^2=8.92181$ ,  $x_B=0.514505$ ,  $t=-16.8609$ ,  $W^2=9.29862$ ,  $y=0.883102$   
 $\theta(l',\gamma)=11.1676$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$l$   $E= 2.3322$   $p_x= -0.2065$   $p_y= 2.0968$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= 0.2065$   $p_y= -2.0968$   $p_z= 0.0000$   
 $l'$   $E= 1.4247$   $p_x= 0.0980$   $p_y= -1.0101$   $p_z= 0.0000$   
 $P'$   $E= 2.2138$   $p_x= 0.0000$   $p_y= 2.0053$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= -0.0980$   $p_y= -0.9952$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=6, retained=100.0%)

